

Relational Reasoning: What We Know and Why It Matters

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Abstract

This article addresses two goals. First, it considers the nature and importance of relational reasoning, the ability to discern meaningful patterns within informational streams. Second, it examines four principles about relational reasoning derived from the empirical literature. Specifically, we argue that relational reasoning is foundational and pervasive; varies as a consequence of age, domain, and context; can be measured in diverse ways; and is malleable and teachable. Evidence supports each of these principles, and the implications for educational policies and practices are weighed.

Keywords

relational reasoning, critical thinking, higher order thinking, analytical thinking, analogical reasoning

Tweet

Without relational reasoning—the ability to discern patterns in information—there can be no critical, analytical, or higher-order thinking.

Key Points

- Relational reasoning is seeing meaningful patterns in information, and can be done using analogy, anomaly, antimony, and antithesis.
- The ability to reason relationally is foundational and pervasive.
- Relational reasoning varies by age, domain, and context.
- Relational reasoning can be measured and observed in diverse ways.
- Relational reasoning is malleable and teachable.
- Critical, analytical, and higher order reasoning all require relational reasoning.

Introduction

A number of recent educational initiatives are designed to improve the quality and depth of students' thinking. These initiatives, including the Common Core State Standards (National Governors Association Center for Best Practices & Council of Chief State School Officers, 2010) and the Next Generation Science Standards (NGSS; 2013), emphasize the need for critical, analytical, and higher order thinking. For example, as expressly stated on the website introducing the

Common Core Standards for English Language Arts, these standards “stress critical-thinking, problem-solving, and analytical skills that are required for success in college, career, and life” (Common Core English Language Arts Standards, 2011). Similarly, the NGSS emphasizes “the critical thinking and communication skills that students need for postsecondary success and citizenship in a world fueled by innovations in science and technology” (NGSS, 2013). Indeed, the notions of critical, analytical, or higher order thinking appear to be driving forces behind widespread campaigns to expand the depth and breadth of students' learning in such areas as English, mathematics, and science (Murphy, Wilkinson, Soter, Hennessey, & Alexander, 2009).

In spite of the value placed in the aforementioned cognitive capacities, the meaning of critical, analytical, or higher order thinking has little conceptual clarity. Moreover, what manner of reasoning or processing might underlie these “higher” forms of human thought is even less understood (Alexander & Disciplined Reading & Learning Research Laboratory [DRLRL], 2012). In effect, if higher forms of human thought are the desired *ends* of the educational enterprise, what are the cognitive *means* by which these ends might be attained? Thus, it would seem essential to delve

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deeper into the very constructs and associated processes that are at the root of this drive toward more critical-analytical thought. For these reasons, our overarching goal is to consider the role relational reasoning plays in all manner of higher thought and, consequently, its place in educational policies and practices.

What Is Relational Reasoning?

The ability to detect meaningful patterns within streams of information lies at the heart of thinking “higher” or “deeper.” This capacity, known as relational reasoning, involves discerning relations among objects, ideas, or situations (Alexander & DRLRL, 2012). This capacity to perceive patterns within a deluge of information and conceptually to map higher order relations onto those perceptions is a precondition for higher order thinking. For instance, when students describe the heart “as a pump,” or recognize correspondences between the wars in Vietnam and Afghanistan, they are reasoning analogically. Specifically, *analogical reasoning*, long considered a centerpiece of higher order thinking, involves recognizing similarities between seemingly dissimilar objects, ideas, or situations (Richland, Zur, & Holyoak, 2007; White & Alexander, 1986).

In addition to analogy, Alexander and colleagues (2012) have empirically established three other forms of relational reasoning that warrant attention: anomaly, antinomy, and antithesis (Dumas, Alexander, Baker, Jablansky, & Dunbar, 2014; Jablansky, Alexander, Dumas, & Compton, 2015). Like analogical reasoning, *anomalous reasoning* not only involves discerning a pattern among a set of objects, ideas, or situations, but it also entails detecting a break from that pattern. Specifically, anomalies represent a deviation from an expected rule (Dumas, Alexander, & Grossnickle, 2013; Klahr & Dunbar, 1988). Within the cognitive literature, anomalous reasoning has been associated with identifying inconsistencies in thinking, and as a catalyst for conceptual change (diSessa, 2002; Vosniadou & Mason, 2012).

By comparison, *antinomies* represent instantiations of incompatibility, that is, the establishment of the criteria for inclusion or exclusion (Alexander & DRLRL, 2012). For example, developmental psychology classifies some models as continuous (Vygotsky, 1962) and others as discontinuous (Piaget, 1928). Thus, one could reason antinomously that it would be impossible to categorize a developmental theory as both continuous and discontinuous. Both anomalies and antinomies are frequently used in scientific contexts, either to detect aberrations or outliers within a set of data (Dumas et al., 2014) or to divide objects or phenomena into ontological categories (Slotta & Chi, 2006).

Finally, *antithetical reasoning* demands the recognition of relational opposites or two ends of a continuum. In an educational context, such opposition may be exemplified by refutational texts (Rapp & Kendeou, 2009) or texts designed to present two viewpoints while systematically dismantling one

(Broughton, Sinatra, & Reynolds, 2009). Overall, antithetical thinking necessitates that students recognize and reconcile conflicting perspectives or sources of evidence—key components of higher order thinking (Alexander et al., 2012).

These four forms of relational reasoning, while individually of significance, often operate in concert (Dumas et al., 2014), and individuals who can effectively use relational reasoning stand a much better chance of achieving depth and breadth of thinking. Relational reasoning entails the crucial and intentional interplay between percepts and concepts that frame all manner of nonintuitive, reflective thought (Alexander & Baggetta, 2014; Stanovich, West, & Toplak, 2011). Given that mental engagement with ideas is a hallmark of higher order thinking (Alexander & DRLRL, 2012), relational reasoning—an effortful and deliberate strategy for relating objects, ideas, or situations—seems a reasonable mechanism for achieving such thinking.

What Is Known About Relational Reasoning?

To further ground our claim that relational reasoning is crucial for “higher” thinking, four principles represent what science understands about this essential cognitive capability:

- The ability to reason relationally is foundational and pervasive.
- Relational reasoning varies by age, domain, and context.
- Relational reasoning can be measured and observed in diverse ways.
- Relational reasoning is malleable and teachable.

The Ability to Reason Relationally Is Foundational and Pervasive

As we (Alexander & DRLRL, 2012) and others (James, 1890; Spearman, 1927) have argued, the ability to discern patterns within any information stream is foundational to learning and performance. For example, in his classic volume, *Principles of Psychology* (1890), William James described the mental ability used to uncover relations of differences and similarity between mental representations. James argued that without such ability, humans would be trapped in a world filled with isolated stimuli, unable to connect objects perceived across time and space. Similarly, the research on perception by Gestalt psychologists demonstrated humans’ tendency to impose some order or structure on otherwise unrelated information (Köhler, 1929; Wertheimer, 1900).

Moreover, contemporary advances in cognitive and neuroscience research further establish relational reasoning’s foundational nature (Bazargani, Hillebrandt, Christoff, & Dumontheil, 2014; Krawczyk, 2012). For example, functional brain imaging enabled neuroscientists to examine relational

reasoning. As such, a growing literature studies higher order processes, such as deductive reasoning (Monti, Osherson, Martinez, & Parsons, 2007) and complex problem solving (Campitelli, Gobet, Williams, & Parker, 2007; Goel & Grafman, 1995). The prefrontal cortex is activated when one reasons relationally (Robin & Holyoak, 1995).

More specifically, the prefrontal cortex has been implicated in the performance of higher level cognitive functions (Duncan, Burgess, & Emslie, 1995; Smith, Patalano, & Jonides, 1998). Neuroscientists hypothesize that given its increase in size over human evolution (Benson, 1993), the prefrontal cortex may be the locus for the system of relational reasoning. Furthermore, these researchers conjecture that its growth may reflect the increasing cognitive demands humans encounter daily (Lenhart, 2005; Purcell, Weigand, & Schall, 2012; Robin & Holyoak, 1995). In effect, these emerging findings in cognition and neuroscience give credence to our claim that relational reason ability is foundational to living and learning in contemporary society.

Relational reasoning is not only foundational but also pervasive, permeating all manner of human thought. More specifically, relational reasoning has been empirically linked to academic achievement in a variety of domains, such as reading (Ehri, Satlow, & Gaskins, 2009), chemistry (Bellocchi & Ritchie, 2011), mathematics (Richland & McDonough, 2010), technological literacy (Jablansky et al., 2015), and nursing education (Fountain, 2015). Beyond academic domains, the pervasiveness of relational reasoning has been repeatedly demonstrated in the words and actions of professionals engaged in critical and creative problem solving (Dumas et al., 2014; Paletz, Schunn, & Kim, 2013).

For example, a study exploring relational reasoning in medical education found all four forms of relational reasoning during discussions between an attending clinical neurologist and his team of residents making diagnostic and therapeutic decisions about their patients in a hospital setting (Dumas et al., 2014). Another example of the ubiquity of relational reasoning in real-world contexts (Dumas & Schmidt, 2015) examined relational reasoning as a predictor of the quantity and quality of engineering designs; relational reasoning was an underlying ability that supported the innovative processes required for the production of creative solutions to mechanical problems. Based on these examples, relational reasoning is a cognitive capacity required in the execution of complex tasks—whether in schools or in professional venues.

Relational Reasoning Varies by Age, Domain, and Context

Internal and external factors ultimately determine the quality and quantity of relational reasoning. These factors include reasoners' age, knowledge of, and familiarity with the domain and task at hand, as well as the reasoning context.

Clearly, relational reasoning operates across the life span, observed in children as young as four (Gentner, 1977; White & Alexander, 1986) and adults older than 80 (Fischer, Norberg, & Lundman, 2008). Although most research on young children centers on analogical reasoning (e.g., Gentner, Loewenstein, & Hung, 2007), a cross-sectional study of kindergarten through 11th-grade students found analogy, anomaly, antinomy, and antithesis in their discourse at all grade levels (Jablansky et al., 2015). The relative prevalence for each form, however, differed by age. Specifically, analogical and anomalous reasoning prevailed in the talk of children between kindergarten and fourth grade, while children in sixth to 11th grade demonstrated more antinomous and antithetical reasoning (Jablansky et al., 2015).

These results suggest differentiating between “typical” and “maximal” performance (Dubois, Sackett, Zedeck, & Fogli, 1993). Although children can reason relationally, their “default” forms of relational reasoning differ from those that can be prompted under certain circumstances. Contextual factors, such as the person or object that elicits the reasoning, are therefore key in determining the occasions and forms that are ultimately expressed. For example (Jablansky et al., 2015), students' verbalized forms of relational reasoning appeared to be shaped, in part, by support offered by the researcher whose questions and comments scaffolded students' thinking. When such supports are minimal, the relational reasoning manifested might well be constrained (e.g., Taverna & Peralta, 2013). Moreover, the familiarity with the object or topic, along with prior learning opportunities (Klahr & Dunbar, 1988), may influence the occasions of relational reasoning (Defeyter, Hearing, & German, 2009; Jablansky et al., 2015). In this way, age necessarily interacts with environmental supports and individuals' prior experiences with the objects of reasoning.

Beyond age, the domain per se can affect the forms of relational reasoning. For instance, medicine demands anomalous reasoning to identify abnormalities in a patient's symptoms (Dumas et al., 2014). In contrast, writing can call for antithetical reasoning when synthesizing across multiple sources (Diakidoy, Mouskounti, & Ioannides, 2011). However, a domain does not elicit only one type of relational reasoning. As a case in point, a reasoning sequence emerged, as doctors analyzed patients' symptoms and worked toward diagnoses. While doctors generally began by identifying a case's anomalies, they often followed with antinomies, and then analogical references to previous cases, before making a preliminary diagnosis (Dumas et al., 2014).

Tasks themselves contribute to relational reasoning's manifestations. When the task involves classifying or differentiating among categories of objects, antinomous reasoning is most prevalent, be the domain physics (Schneider & Hardy, 2013), technology (Jablansky et al., 2015), or engineering (Dumas & Schmidt, 2015). By contrast, when the task involves reconciling conflicting sources of information,

antithetical reasoning may prove most useful (Hynd & Alvermann, 1986; van den Broek & Kendeou, 2008). Relational reasoning is a multifaceted construct whose manifestations are multiply determined. Verbalizations of each form are influenced by age and domain, with the contexts of task, environment, and experience contributing.

Relational Reasoning Can Be Observed and Measured in Diverse Ways

Given that relational reasoning manifests differently as a function of individual and environmental factors, the construct itself can be measured in a variety of ways. Educational and psychological researchers interested in complex thinking and learning use diverse measures and methodologies for tapping into relational reasoning ability. That variety includes psychometric measures and tasks (e.g., Alexander, Dumas, et al., 2015; Richland, Chan, Morrison, & Au, 2010), naturalistic observations (e.g., Chan & Schunn, 2015; Dumas et al., 2014), and cognitive interview techniques (Jablansky et al., 2015).

Of the possible methods for assessing relational reasoning, one in particular has stood out: four-term analogies (Alexander, White, Haensly, & Crimmins-Jeanes, 1987; Goswami & Bryant, 1992; Maguire, McClelland, Donovan, Tillman, & Krawczyk, 2012; Sternberg, 1977). Four-term analogies follow the form A:B::C:D, where the A and B term are linked to the C and D term by a higher order relation of similarity. For example, the four-term verbal analogy *wolf:pack::lion:pride* allows for the identification of the lower order relation of “is a member of” between *wolf* and *pack* as well as *lion* and *pride*. A higher order relation of similarity links the two lower order relations, forming an analogy. Four-term analogies do not necessarily need to be verbal in nature, but can be presented pictorially (e.g., Krawczyk et al., 2008; Richland et al., 2010) or figurally (e.g., Tunteler & Resing, 2010).

One commonly used expansion of classic analogy problems is matrix analogies (Alexander et al., 2015; Naglieri & Insko, 1986; Raven, 1941). Today, the matrix format remains the gold standard for assessing visuospatial analogical reasoning and fluid ability (Krawczyk, McClelland, & Donovan, 2011). Matrix analogies are also the measure of choice for many neuroscientists working in relational reasoning, perhaps because their complex visuospatial nature suit them for neuroimaging brain activation (see Krawczyk, 2012, for a review). Recently, Alexander and colleagues (2012) have built on the matrix format to assess other forms of relational reasoning (e.g., antinomy). Together, these figural arrays constitute the Test of Relational Reasoning (TORR; Alexander, 2012; Dumas & Alexander, in press).

While the previously described measures utilize a selected-response (closed-ended) format, open-ended and naturalistic observations have also tapped relational reasoning. For example, text-based studies, especially those

incorporating refutational texts, challenge participants to identify and describe higher order relations while reading (Kendeou & O’Brien, 2015; Sinatra & Broughton, 2011). Moreover, cognitive interviews can capture students’ relational reasoning about particular objects or processes (e.g., Jablansky et al., 2015; Rush & Roy, 2001). Semistructured interviews, in which the researcher prompts or supports participants’ relational reasoning, may be especially useful for young respondents (e.g., Jablansky et al., 2015). Furthermore, naturalistic, or in vivo methodologies, have repeatedly captured relational reasoning on complex problems (e.g., Chan & Schunn, 2015; Dumas et al., 2014; Trickett, Trafton, & Schunn, 2009). These methods, which tap relational reasoning as it occurs within a domain-specific problem-solving context, may be necessary to uncover differences related to domains (e.g., engineering or medicine) or levels of expertise (e.g., teacher or student; attending or resident physician), or to model how various reasoning forms work in concert in educational or professional contexts.

Based on extant investigations, the relational reasoning construct still has much remaining to be learned. For greater insights, the entire gamut of measurement tools must be used: from selected-response assessments and neuroimaging methods to semistructured interviews and naturalistic observations. In this way, the nature of relational reasoning may be more fully uncovered, and those engaged in complex reasoning in school and the workplace may be systematically supported.

Relational Reasoning Is Malleable and Teachable

Relational reasoning capability should not be viewed as a predetermined or fixed entity (Blackwell, Trzesniewski, & Dweck, 2007; Dweck & Bempechat, 1983; Mangels, Butterfield, Lamb, Good, & Dweck, 2006). Rather, an individual’s ability to discern meaningful patterns and to utilize that capacity strategically can be shaped through relevant experiences. Drawing analogously on the empirical work in intelligence, the simple understanding that relational reasoning can transform to some degree can motivate and catalyze cognitive and metacognitive action (Burke & Williams, 2012; Cornoldi, 2010; Miele, Son, & Metcalfe, 2013).

The events that might shape relational reasoning can be both direct and indirect. For instance (Murphy et al., 2016), a brief compare–contrast writing intervention affected students’ relational reasoning in a fifth-grade classroom: students’ ability to engage in relational reasoning and generate, compare, and reconcile arguments developed in line with the writing intervention. Similarly, high-school physics and chemistry students’ relational reasoning and scientific understanding gained through an intervention that focused on evidence-based discussions (Greene et al., 2016).

In an example of the impact of a more indirect intervention on relational reasoning ability (Dumas & Schmidt,

2015), graduate students in mechanical engineering received inventive problem-solving instruction (i.e., TRIZ) to improve solutions to design problems. While relational reasoning predicted ideation generation success prior to the TRIZ, the association between TRIZ and relational reasoning strengthened after the intervention. In sum, if relational reasoning were a fixed construct, none of the changes in relational reasoning ability described herein would manifest.

In the aforementioned cases, relational reasoning was broadly targeted through various experiences such as compare–contrast lessons (Murphy et al., 2016) or a creative problem-solving methodology (Dumas & Schmidt, 2015). Component-based analogy training studies, grounded in the processes underlying thinking (i.e., encoding, inferring, mapping, and applying; Sternberg, 1982), offer additional credence to the claim that relational reasoning is malleable and teachable. For instance, explicit instruction of the component processes (e.g., inferring and mapping) facilitate elementary, middle, and secondary school students' ability to solve analogy problems in linguistic and nonlinguistic forms and in the domains of history and biology (Alexander, Pate, Kulikowich, Farrell, & Wright, 1989; Alexander, White, & Mangano, 1983). Furthermore, late adolescents and young adults provide evidence of successful analogy performance following component-based training (Sternberg & Ketron, 1982).

The role of the same components to the processing of anomalous, antinomial, and antithetical forms of relational reasoning has also been empirically established (Grossnickle, Dumas, Alexander, & Baggetta, 2015). Thus, as the prior studies suggest, the component processes underlying relational reasoning are ripe for direct intervention. Specifically, this ability to discern meaningful patterns within any information stream seemingly relies on basic cognitive processes that already have a place within learning environments across grades and academic domains (Ward, Suss, & Basevitch, 2009). The goal should be to make these component processes more transparent and to demonstrate how they uniquely combine to give each form of relational reasoning its defining character. Of course, much is still to be learned about the best time, place, and contexts to enhance relational reasoning. Yet, if relational reasoning is demonstrably malleable and teachable, then to ignore those educational opportunities seems unconscionable.

The Implications of Relational Reasoning for Policies and Practices

The capability to discern meaningful patterns in the flow of information, whether in school or in the world at large, is pivotal to learning and development. As William James (1890) contended,

Our first way of looking at a reality is often to suppose it simple, but later we may learn to perceive it as compound. This new way

of knowing the same reality may conveniently be called by the name of Analysis. It is manifestly one of the most incessantly performed of all our mental processes. (p. 502)

Without this capacity to see beyond the simple, via analysis of similarities and differences (i.e., reasoning relationally), there can be no depth of learning, no academic development. Yet, beyond its unquestioned potency to undergird human perceptions and conceptions and to give rise to thoughts that are defensibly higher order in nature, researchers' and educators' knowledge and harnessing of relational reasoning remain in their infancy.

To address the growing literature on the interface between relational reasoning and learning and development, several empirically defensible principles hold promise for preparing today's learners to confront the deluge of information that they unceasingly encounter in school and out of school. But, how do we translate these principles of relational reasoning into actionable and sound decisions about how to teach, what to teach, and what to assess? Such concerns must be thoughtfully weighed; for without such translations of theory and research into educational policies and practices, insights into relational reasoning will remain no more than inert knowledge (Whitehead, 1929). Thus, we offer several "actions" that merit due consideration by those responsible for shaping the educational agenda and learning environments for today's and tomorrow's students.

Think Critically and Analytically About Higher Order Thinking

The contemporary focus on critical, analytical, and higher order thinking in educational policies and curricular initiatives is laudable. Yet, these policies and initiatives demand deeper reflection on the processes and components that ultimately fuel these alternatives to superficial thinking (Stanovich et al., 2011). Relational reasoning can serve that goal. It can help promote the systematic attention to similarities and dissimilarities in what is seen, heard, or sensed. Relational reasoning promotes examining not just analogical or similarity associations but also associations of discrepancy, exclusion, and opposition. It can be the mechanism by which ideas, objects, or situations are perceived in conjunction rather than as isolated phenomena. Relational reasoning can, therefore, become a means to those worthwhile ends of critical, analytical, and higher order thought.

Unearth the Potential of Relational Reasoning

It is not enough to recognize the foundational character of relational reasoning or acknowledge its central role in all forms of "higher" thinking. What is also essential is to devise measures and techniques that would allow researchers and practitioners to make the internal mechanisms of relational

reasoning more transparent. Until students' relational reasoning can be made visible, there will be no way to document its role in academic achievement or to permit students to bear witness to the power of thinking analogically, anomalously, antinomously, or antithetically. Toward that end, we have been laboring to devise psychometrically sound measures of relational reasoning that be used with elementary and middle school students (i.e., TORRjr; Jablansky, Alexander, & Singer, 2016). We have also developed and calibrated a figural measure of relational reasoning for older students and adults (TORR; Alexander, 2012; Alexander et al., 2015), as well as a verbal form for this same population (vTORR; Alexander, Singer, Jablansky, & Hattan, 2015).

These formal measures assess students' capacity to reason relationally in a manner not previously available. These formal measures complement with creative techniques for documenting relational reasoning within the flow of naturally occurring problem solving, including among medical professionals and children engaged in technical literacy tasks (Dumas et al., 2014; Jablansky et al., 2015). Collectively these in vivo techniques, combined with more formal assessment tools, can afford researchers and practitioners the means of excavating the reasoning processes of individuals at all ages in diverse contexts.

Be Models of the Processes of Relational Reasoning

It would seem unjust to expect students to demonstrate the depth of thinking fueled by relational reasoning, without populating classrooms with teachers model these valued mental processes. Yet, given that relational reasoning is apt to be a novel, unfamiliar notion within teacher education, we cannot presume that those guiding the learning of others are prepared to assume the role of reasoning model. That is not to say that teachers are incapable of rising to that task if they are provided with the professional development required. These professional development initiatives should not attempt simply to lay relational reasoning on top of all other agendas and mandates that fall on teachers' shoulders. Rather, professional development should strive to demonstrate how relational reasoning could be woven into the existing fabric of the educational experience. Teachers and administrators can see that relational reasoning could empower the agendas and mandates already in place—not simply add to them.

Arm Students With the Language and Knowledge of Relational Reasoning

Finally, our concluding principle sought to make clear that relational reasoning is by no means a fixed capacity. Rather, relational reasoning is susceptible to experiences that stimulate students' perceptiveness, demand the integration of ideas, and necessitate the resolution of contrasting evidence.

If relational reasoning is both foundational and teachable, then we have no choice but to make such transformational experiences available to students. An initial step in these experiences would be to equip students with the language of relational reasoning. Teach them what analogy, anomaly, antinomy, and antithesis mean, and encourage students to use those terms when engaged in the analysis of ideas, objects, and situations.

Beyond labeling relational reasoning forms (e.g., antinomy and antithesis) and underlying component processes (i.e., mapping and applying; Sternberg, 1977), students need to witness firsthand the power of relational reasoning across academic domains and tasks. In effect, as students study reading, history, mathematics, or science, they should be given the opportunity to craft analogies, identify anomalies, point out antinomies, and forge antitheses. Furthermore, students should be helped to realize how these reasoning processes are central to all academic domains and to see how successful readers, historians, mathematicians, and scientists are ultimately successful relational reasoners. Moreover, the assessment of learning within these academic domains should require manifesting these foundational reasoning processes. Aligning foundational capacities with effective instructional initiatives and innovative assessment practices will help make the praiseworthy goal of an informed citizenry a reality.

Clearly, these four recommendations are modest actions that should be undertaken within the educational enterprise. Specifically, teachers and administrators can (a) use relational reasoning theory and research to help refine current notions of "higher" thinking, (b) utilize measures and methods to bring students' relational reasoning to the surface for inspection and reshaping, (c) model the forms and processes of relational reasoning within the learning environment, (d) teach the forms and component processes of relational reasoning, and (e) make those processes central to assessment practices. To do otherwise would be to let the instructional power of relational reasoning wither; to do otherwise would be to watch students' reasoning potential remain unfulfilled; and to do otherwise would be to miss the opportunity to strengthen existing policies that address critical, analytical, and higher order thinking.

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References

- Alexander, P. A. (2012). *The Test of Relational Reasoning*. College Park, MD: Disciplined Reading and Learning Research Laboratory.

- Alexander, P. A., & Baggetta, P. (2014). Percept-concept coupling and human error. In D. N. Rapp & J. L. G. Braasch (Eds.), *Processing inaccurate information: Theoretical and applied perspectives from cognitive science and the educational sciences* (pp. 297-328). Cambridge, MA: MIT Press.
- Alexander, P. A., & Disciplined Reading and Learning Research Laboratory. (2012). Reading into the future: Competence for the 21st century. *Educational Psychologist*, 47, 259-280.
- Alexander, P. A., Dumas, D., Grossnickle, E. M., List, A., & Firetto, C. (2015). Measuring relational reasoning. *Journal of Experimental Education*, 83, 1-33.
- Alexander, P. A., Pate, P. E., Kulikowich, J. M., Farrell, D. M., & Wright, N. L. (1989). Domain-specific and strategic knowledge: Effects of training on students of differing ages or competence levels. *Learning and Individual Differences*, 1, 283-325.
- Alexander, P. A., Singer, L. M., Jablansky, S., & Hattan, C. (2015). Relational reasoning in word and in figure. *Journal of Educational Psychology*. Advance online publication.
- Alexander, P. A., White, C. S., Haensly, P. A., & Crimmins-Jeanes, M. (1987). Training in analogical reasoning. *American Educational Research Journal*, 24, 387-404.
- Alexander, P. A., White, C. S., & Mangano, N. (1983). Examining the effects of direct instruction in analogical reasoning on reading comprehension. In J. A. Niles & L. A. Harris (Eds.), *Searches in meaning in reading/language processing and instruction: Thirty-second yearbook of the National Reading Conference* (pp. 36-41). Rochester, NY: National Reading Conference.
- Bazargani, N., Hillebrandt, H., Christoff, K., & Dumontheil, I. (2014). Developmental changes in effective connectivity associated with relational reasoning. *Human Brain Mapping*, 35, 3262-3276.
- Bellocchi, A., & Ritchie, S. M. (2011). Investigating and theorizing discourse during analogy writing in chemistry. *Journal of Research in Science Teaching*, 48, 771-792.
- Benson, D. F. (1993). Prefrontal abilities. *Behavioural Neurology*, 6(2), 75-81.
- Blackwell, L. S., Trzesniewski, K. H., & Dweck, C. S. (2007). Implicit theories of intelligence predict achievement across an adolescent transition: A longitudinal study and an intervention. *Child Development*, 78, 246-263.
- Broughton, S. H., Sinatra, G. M., & Reynolds, R. E. (2009). Attention allocation and the refutation text effect: A possible process mechanism for conceptual change. In G. M. Sinatra (Chair), *Mechanisms for facilitating conceptual change through text-based interventions*. Symposium presented at the annual meeting of the American Educational Researchers Association, San Diego, CA.
- Burke, L. A., & Williams, J. M. (2012). Two thinking skills assessment approaches: "Assessment of pupils' thinking skills" and "individual thinking skills assessments." *Thinking Skills and Creativity*, 7(1), 62-68.
- Campitelli, G., Gobet, F., Williams, G., & Parker, A. (2007). Integration of perceptual input and visual imagery in chess players: Evidence from eye movements. *Swiss Journal of Psychology*, 66, 201-213.
- Chan, J., & Schunn, C. (2015). The impact of analogies on creative concept generation: Lessons from an in vivo study in engineering design. *Cognitive Science*, 39, 126-155.
- Common Core English Language Arts Standards. (2011). Retrieved from <http://www.corestandards.org/ELA-Literacy/>
- Cornoldi, C. (2010). Metacognition, intelligence, and academic performance. In H. S. Waters & W. Schneider (Eds.), *Metacognition, strategy use, and instruction* (pp. 257-277). New York, NY: Guilford Press.
- Defeyter, M. A., Hearing, J., & German, T. C. (2009). A developmental dissociation between category and function judgments about novel artifacts. *Cognition*, 110, 260-264.
- Diakidoy, I. A. N., Mouskounti, T., & Ioannides, C. (2011). Comprehension and learning from refutation and expository texts. *Reading Research Quarterly*, 46, 22-38.
- diSessa, A. A. (2002). Why "conceptual ecology" is a good idea. In M. Limón & L. Mason (Eds.), *Reconsidering conceptual change: Issues in theory and practice* (pp. 29-60). Dordrecht, The Netherlands: Kluwer Academic.
- Dubois, C. L. Z., Sackett, P. R., Zedeck, S., & Fogli, L. (1993). Further exploration of typical and maximum performance criteria: Definitional issues, prediction, and white-black differences. *Journal of Applied Psychology*, 78, 205-211.
- Dumas, D., & Alexander, P. A. (in press). Calibration of the Test of Relational Reasoning. *Psychological Assessment*.
- Dumas, D., Alexander, P. A., Baker, L. M., Jablansky, S., & Dunbar, K. N. (2014). Relational reasoning in medical education: Patterns in discourse and diagnosis. *Journal of Educational Psychology*, 106, 1021-1035.
- Dumas, D., Alexander, P. A., & Grossnickle, E. M. (2013). Relational reasoning and its manifestations in the educational context: A systematic review of the literature. *Educational Psychology Review*, 25, 391-427.
- Dumas, D., & Schmidt, L. (2015). Relational reasoning as predictor for engineering ideation success using TRIZ. *Journal of Engineering Design*, 26(1-3), 74-88.
- Duncan, J., Burgess, P., & Emslie, H. (1995). Fluid intelligence after frontal lobe lesions. *Neuropsychologia*, 33, 261-268.
- Dweck, C., & Bempechat, J. (1983). Children's theories of intelligence. In S. Paris, G. Olsen, & H. W. Stevenson (Eds.), *Learning and motivation in the classroom* (pp. 239-256). Hillsdale, NJ: Lawrence Erlbaum.
- Ehri, L. C., Satlow, E., & Gaskins, I. (2009). Grapho-phonemic enrichment strengthens keyword analogy instruction for struggling young readers. *Reading & Writing Quarterly*, 25, 162-191.
- Fischer, R. S., Norberg, A., & Lundman, B. (2008). Embracing opposites: Meanings of growing old as narrated by people aged 85. *The International Journal of Aging & Human Development*, 67, 259-271.
- Fountain, L. (2015). *Relations among topic knowledge, individual interest, relational reasoning, and critical thinking in maternity nurses* (Doctoral dissertation proposal). University of Maryland, College Park.
- Gentner, D. (1977). Children's performance on a spatial analogies task. *Child Development*, 48, 1034-1039.
- Gentner, D., Loewenstein, J., & Hung, B. (2007). Comparison facilitates children's learning of names for parts. *Journal of Cognition and Development*, 8, 285-307.
- Goel, V., & Grafman, J. (1995). Are the frontal lobes implicated in "planning" functions? Interpreting data from the Tower of Hanoi. *Neuropsychologia*, 33, 623-642.

- Goswami, U., & Bryant, P. (1992). Rhyme, analogy, and children's reading. In P. B. Gough, L. C. Ehri, & R. Treiman (Eds.), *Reading acquisition* (pp. 49-63). Hillsdale, NJ: Lawrence Erlbaum.
- Greene, J. A., Murphy, P. K., Butler, A., Firetto, C. M., Allen, E., Wang, J., . . . Yu, S. B. (2016, April). Fostering relational reasoning and scientific understanding through quality talk discourse. In Dumas D. (Chair), *The malleability of relational reasoning*. Symposium conducted at the annual meeting of the American Educational Research Association, Washington, DC.
- Grossnickle, E., Dumas, D., Alexander, P. A., & Baggetta, P. (2015). *The process of reasoning relationally*. Manuscript under review.
- Hynd, C., & Alvermann, D. E. (1986). The role of refutation text in overcoming difficulty with science concepts. *Journal of Reading*, 29, 440-446.
- Jablansky, S., Alexander, P. A., Dumas, D., & Compton, V. (2015). Developmental differences in relational reasoning among primary and secondary school students. *Journal of Educational Psychology*. Advance online publication.
- Jablansky, S., Alexander, P. A., & Singer, L. (2016, April). *Examining the relational reasoning capabilities of elementary and middle-school students with learning needs*. Paper accepted for presentation at the American Education Research Association conference, Washington, DC.
- James, W. (1890). *The principles of psychology*. New York, NY: Henry Holt.
- Kendeou, P., & O'Brien, N. (2015, April). Antithetical reasoning with refutational texts. In Sinatra G.M. (Chair), *Relational reasoning in STEM domains: What empirical research can contribute to the national dialogue*. Symposium conducted at the annual meeting of the American Educational Research Association, Chicago, IL.
- Klahr, D., & Dunbar, K. (1988). Dual search space during scientific reasoning. *Cognitive Science*, 12, 1-48.
- Köhler, W. (1929). *Gestalt psychology*. New York, NY: Horace Liveright.
- Krawczyk, D. C. (2012). The cognition and neuroscience of relational reasoning. *Brain Research*, 1428, 13-23.
- Krawczyk, D. C., McClelland, M. M., & Donovan, C. M. (2011). A hierarchy for relational reasoning in the prefrontal cortex. *Cortex*, 47, 588-597.
- Krawczyk, D. C., Morrison, R. G., Viskontas, I., Holyoak, K. J., Chow, T. W., Mendez, M. F., . . . Knowlton, B. J. (2008). Distraction during relational reasoning: The role of prefrontal cortex in interference control. *Neuropsychologia*, 46, 2020-2032.
- Lenhart, A. B. (2005). *Unstable texts: An ethnographic look at how bloggers and their audience negotiate self-presentation, authenticity and norm formation* (Doctoral dissertation). Georgetown University, Washington, DC.
- Maguire, M. J., McClelland, M. M., Donovan, C. M., Tillman, G. D., & Krawczyk, D. C. (2012). Tracking cognitive phases in analogical reasoning with event-related potentials. *Journal of Experimental Psychology: Learning, Memory, and Cognition*, 38, 273-281.
- Mangels, J. A., Butterfield, B., Lamb, J., Good, C., & Dweck, C. S. (2006). Why do beliefs about intelligence influence learning success? A social cognitive neuroscience model. *Social Cognitive and Affective Neuroscience*, 1, 75-86.
- Miele, D. B., Son, L. K., & Metcalfe, J. (2013). Children's naive theories of intelligence influence their metacognitive judgments. *Child Development*, 84, 1879-1886.
- Monti, M. M., Osherson, D. N., Martinez, M. J., & Parsons, L. M. (2007). Functional neuroanatomy of deductive inference: A language-independent distributed network. *NeuroImage*, 37, 1005-1016.
- Murphy, P. K., Greene, J. A., Firetto, C. M., Montalbano, C., Mengyi, L., Wei, L., & Croninger, R. M. V. (2016, April). Promoting relational reasoning in elementary students' writing. In Dumas D. (Chair), *The malleability of relational reasoning*. Symposium conducted at the annual meeting of the American Educational Research Association, Washington, DC.
- Murphy, P. K., Wilkinson, I. A. G., Soter, A. O., Hennessey, M. N., & Alexander, J. F. (2009). Examining the effects of classroom discussion on students' high-level comprehension of text: A meta-analysis. *Journal of Educational Psychology*, 101, 740-764.
- Naglieri, J. A., & Insko, W. R. (1986). Construct validity of the Matrix Analogies Test-Expanded Form. *Journal of Psychoeducational Assessment*, 4, 243-255.
- National Governors Association Center for Best Practices & Council of Chief State School Officers (2010). *Common Core State Standards*. Washington, DC: Author.
- Next Generation Science Standards. (2013). Retrieved from <http://www.nextgenscience.org/frequently-asked-questions#2.1>
- Paletz, S. B. F., Schunn, C. D., & Kim, K. (2013). The interplay of conflict and analogy in multidisciplinary teams. *Cognition*, 126, 1-19.
- Piaget, J. (1928). *Judgment and reasoning in the child*. London, UK: Routledge & Kegan Paul.
- Purcell, B. A., Weigand, P. K., & Schall, J. D. (2012). Supplementary eye field during visual search: Saliency, cognitive control, and performance monitoring. *The Journal of Neuroscience*, 32, 10273-10285.
- Rapp, D. N., & Kendeou, P. (2009). Noticing and revising discrepancies as texts unfold. *Discourse Processes*, 46, 1-24.
- Raven, J. C. (1941). Standardization of progressive matrices. *British Journal of Medical Psychology*, 19, 137-150.
- Richland, L. E., Chan, T. K., Morrison, R. G., & Au, T. K. F. (2010). Young children's analogical reasoning across cultures: Similarities and differences. *Journal of Experimental Child Psychology*, 105, 146-153.
- Richland, L. E., & McDonough, I. M. (2010). Learning by analogy: Discriminating between potential analogs. *Contemporary Educational Psychology*, 35, 28-43.
- Richland, L. E., Zur, O., & Holyoak, K. J. (2007). Cognitive supports for analogies in the mathematics classroom. *Science*, 316, 1128-1129.
- Robin, N., & Holyoak, K. J. (1995). Relational complexity and the functions of prefrontal cortex. In M. S. Gazzaniga (Ed.), *The cognitive neurosciences* (pp. 987-997). Cambridge, MA: MIT Press.
- Rush, C., & Roy, R. (2001). Expert judgment in cost estimating: Modelling the reasoning process. *Concurrent Engineering*, 9, 271-284.

- Schneider, M., & Hardy, I. (2013). Profiles of inconsistent knowledge in children's pathways of conceptual change. *Developmental Psychology*, 49, 1639-1649.
- Sinatra, G. M., & Broughton, S. H. (2011). Bridging reading comprehension and conceptual change in science education: The promise of refutation text. *Reading Research Quarterly*, 46, 374-393.
- Slotta, J. D., & Chi, M. T. H. (2006). Helping students understand challenging topics in science through ontology training. *Cognition and Instruction*, 24, 261-289.
- Smith, E. E., Patalano, A. L., & Jonides, J. (1998). Alternative strategies of categorization. *Cognition*, 65, 167-196.
- Spearman, C. (1927). *The abilities of man: Their nature and measurement*. New York, NY: Macmillan.
- Stanovich, K. E., West, R. F., & Toplak, M. E. (2011). Intelligence and rationality. In R. J. Sternberg & S. B. Kaufman (Eds.), *Cambridge handbook of intelligence* (pp. 784-826). New York, NY: Cambridge University Press.
- Sternberg, R. J. (1977). *Intelligence, information processing, and analogical reasoning: The componential analysis of human abilities*. Oxford, UK: Lawrence Erlbaum.
- Sternberg, R. J. (1982). *Handbook of human intelligence*. Cambridge, UK: Cambridge University Press.
- Sternberg, R. J., & Ketron, J. L. (1982). Selection and implementation of strategies in reasoning by analogy. *Journal of Educational Psychology*, 74, 399-413.
- Taverna, A. S., & Peralta, O. A. (2013). Young children category learning: A training study. *European Journal of Psychology of Education*, 28, 599-617.
- Trickett, S. B., Trafton, J. G., & Schunn, C. D. (2009). How do scientists respond to anomalies? Different strategies used in basic and applied science. *Topics in Cognitive Science*, 1, 711-729.
- Tunteler, E., & Resing, W. C. M. (2010). The effects of self and other-scaffolding on progression and variation in children's geometric analogy performance: A microgenetic research. *Journal of Cognitive Education and Psychology*, 9, 251-272.
- van den Broek, P., & Kendeou, P. (2008). Cognitive processes in comprehension of science texts: The role of co-activation in confronting misconceptions. *Applied Cognitive Psychology*, 22, 335-351.
- Vosniadou, S., & Mason, L. (2012). Conceptual change induced by instruction: A complex interplay of multiple factors. In K. Harris, S. Graham, & T. Urdan (Eds.), *Handbook of educational psychology* (Vol. 2, pp. 221-246). Washington, DC: American Psychiatric Association.
- Vygotsky, L. S. (1962). *Thought and language*. Cambridge, MA: MIT Press.
- Ward, P., Suss, J., & Basevitch, I. (2009). Expertise and expert performance-based training (ExPerT) in complex domains. *Technology, Instruction, Cognition and Learning*, 7, 121-145.
- Wertheimer, M. (1900). *Gestalt theory*. Raleigh, NC: Hayes Barton Press.
- White, C. S., & Alexander, P. A. (1986). Effects of training four-year-olds' ability to solve geometric analogy problems. *Cognition and Instruction*, 3, 261-268.
- Whitehead, A. N. (1929). *The aims of education*. New York, NY: The Free Press.